

Cradle-to-gate life cycle assessment of CFRP reinforcement for concrete structures: Calculation basis and exemplary application

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ABSTRACT

The building industry accounts for a significant amount of CO₂ emissions worldwide, at which the building material concrete with its binder cement is responsible for a considerable share. Consequently, it is inevitable to strive for solutions to reduce the amount of concrete used in the building industry. One possible approach is to use reinforcement made of carbon fiber-reinforced polymers (CFRP) combined with appropriate new structural design approaches to reduce the dimensions and thus the weight of concrete building components. CFRP reinforcement is characterized by a higher tensile strength and favorable properties regarding corrosion resistance compared to conventional steel reinforcement. However, the production of carbon fibers is resource- and energy-intensive. This research study evaluates the environmental performance of CFRP reinforcement in concrete structures in accordance with the current state of data availability and standardization. A cradle-to-gate life cycle assessment is conducted. Design values allowing for a preliminary estimation of the environmental performance of CFRP reinforcement are provided, at which maximum transparency regarding the identification process and the consulted environmental data is set as top priority. The addressed impact categories include the abiotic depletion of fossil resources (ADPF), the acidification potential (AP) and the global warming potential (GWP). A comparison of CFRP and conventional steel reinforcement shows the comparatively high environmental impact of CFRP reinforcement amongst all considered impact categories. A subsequent assessment by means of considering the superstructure (deck and railing) of built pedestrian bridge examples including a carbon concrete bridge, a conventional reinforced concrete bridge and a mild steel bridge, indicates otherwise: Due to a significant reduction in self-weight, the GWP and ADPF of the carbon concrete bridge show the comparatively lowest values. These results prove the economic potential of using CFRP as reinforcement based on a quantitative evaluation of the ecologic impact.

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1. Introduction

The building sector accounted for more than one third of anthropogenic carbon dioxide emissions in 2017. This currently upward trend illustrates the urgency for sustainable building solutions as well as more efficient construction material usage. In this context, reinforced concrete is the most widely used construction material in the global non-residential and residential building sector (International Energy Agency, 2018). The production of concrete's indispensable constituent cement represents the third-largest source of anthropogenic carbon dioxide emissions worldwide (Andrew, 2018).

The corollary of this significant environmental impact of cement is to decrease its use and maximize its utilization. Kromoser et al. (2018) introduced three approaches to decrease the environmental impact of concrete structures. One possible approach is to optimize the geometry of structures in accordance with the properties of the used materials. For example, by the use of topology optimization cement consumption can be decreased due to a targeted material reduction in concrete structures. To a certain extent, additives like fly ash or slag sand can also replace the hydraulic binder cement. A further approach is to use high-performance materials in concrete construction. Their high strength and stiffness properties allow for a significant reduction in concrete consumption. In recent years, fiber-reinforced polymer (short: FRP) reinforcement as an alternative to steel reinforcement attracted great attention in research as well as in the construction sector. Ehlig et al. (2012) for example give an overview of built textile-

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Acronyms			
ADP_f	Abiotic Depletion Potential: fossil fuels	EPD	Environmental Product Declaration
AP	Acidification Potential	EPLCA	European Platform on Life Cycle Assessment
B1-CCB	Bridge example 1: carbon concrete bridge	FEICA	Association of the European Adhesive & Sealant Industry
B2-RCB	Bridge example 2: reinforced concrete bridge	FRP	Fiber-Reinforced Polymer
B3-SB	Bridge example 3: mild steel bridge	GWP	Global Warming Potential
BF	Blast Furnace	IBU	Institut Bauen und Umwelt e.V.
BFRP	Basalt Fiber-Reinforced Polymer	LCA	Life Cycle Assessment
BOS	Basic Oxygen Steelmaking	LCI	Life Cycle Inventory
CFRP	Carbon Fiber-Reinforced Polymer	LCIA	Life Cycle Impact Assessment
DU	Declared Unit	LER	Liquid Epoxy Resin
EAF	Electric Arc Furnace	ODP	Ozone Depletion Potential
EIA	Environmental Impact Assessment	PAN	Polyacrylonitrile
ELCD	European reference Life Cycle Database	UHPC	Ultra-High Performance Concrete
		UHPFRC	Ultra-High Performance Fiber-Reinforced Concrete

reinforced concrete structures. Construction components with FRP reinforcement show high potential to build lightweight, durable concrete structures (Kromoser et al., 2018). Unlike other fiber types like basalt, glass and aramid, carbon fiber-reinforced polymer (short: CFRP) is characterized by a high Young's modulus equivalent to the one of steel. This favorable stiffness property makes CFRP reinforcement the favored choice when keeping building structures' deflections to a minimum as required in the serviceability limit state. Further advantages of CFRP reinforcement include a high tensile strength of up to 3000 MPa and above as well as a distinctive non-corrosive behavior. A concrete cover minimum of 10 mm or less becomes feasible, which results in a reduction of material usage in concrete structures.

Besides the mentioned advantages, a high crude oil-based raw material and energy demand in the production phase of CFRP reinforcement constitutes one of the most significant downsides in comparison to steel reinforcement as its conventional pendant. In this context, design values allowing for an estimation of the environmental performance of CFRP reinforcement and consequently CFRP reinforced structures are rarely available. Conducted studies on this subject include the study of Inman et al. (2017), with basalt fiber-reinforced polymer (short: BFRP) reinforcement in concrete beams as matter of investigation. The results show the environmental potential of BFRP reinforcement, with its drawback of less favorable stiffness properties in comparison to steel reinforcement. Furthermore, Maxineasa et al. (2015) conducted an environmental assessment of CFRP as post-strengthening reinforcement in comparison to the demolition and reconstruction of reinforced concrete beams. The study was carried out using GaBi 6 as life cycle assessment (short: LCA) software with results in favor of CFRP strengthened concrete structures regarding global warming, human toxicity and ozone depletion potential.

The studies listed above hardly allow for reproducibility nor give detailed insight into the materials' environmental data. Beyond that, analyses of updated environmental data are still necessary to establish deeper understanding of the environmental performance of FRP reinforcement (Zhang, 2014). This study provides a transparent analysis of the environmental performance of CFRP reinforcement. It intends to serve as a basis for calculation and decisions regarding future research and applications of CFRP reinforcement and CFRP reinforced concrete structures.

The scope of this article includes a cradle-to-gate LCA of CFRP reinforcement and a comparison to conventional steel reinforcement. Subsequently, the environmental performance of CFRP reinforcement is assessed by means of built examples with comparable design properties. A CFRP reinforced concrete bridge is

compared to a conventionally reinforced as well as a mild steel bridge. The calculations are performed in compliance with underlying standardization, which is outlined within the theoretical framework. The derived results of three impact categories are presented. These impact categories include the abiotic depletion potential of fossil fuels, the acidification potential for soil and water and the global warming potential. A discussion of the results as well as recommendations for future research complete the study.

2. Theoretical framework

This section outlines the theoretical framework relevant for the subsequent environmental assessment of CFRP reinforcement.

2.1. LCA fundamentals

Concrete structures typically cause environmental degradation, at which various stages of their life cycle show different environmental impact. Stages of that kind include the production, use or disposal phase. An evaluation of these life cycle phases is fundamental to understand the product's environmental performance and to allow for comparability as well as targeted actions towards a decrease of the environmental impact. According to ISO 14040 (2006), LCA is defined as a "compilation and evaluation of the inputs, outputs and the potential environmental impacts of a product system throughout its life cycle". Life cycle inventory (short: LCI) and life cycle impact assessment (short: LCIA) are consecutive parts of an LCA. LCI aims at collecting and analyzing environmental interventions data associated with a product (emissions, resource consumption, waste generation, etc.). LCIA constitutes the estimation of indicators of environmental pressures, e.g. climate change, smog, resource depletion, acidification and human health effects. Characterization factors are applied within this process to convert LCI data to environmental indicators.

LCA assists in developing standardized evaluation tools like an environmental product declaration (short: EPD), which provides quantified environmentally related data and enables comparison between different products of the same functionality. An EPD can be used as a documentation to demonstrate that harmonized rules for the marketing of construction products after the Construction Products Regulation of the European Commission are met. It can also be consulted as a database for building rating tools. EPDs of construction products are regulated in ISO 14025 (2006) and ÖNORM EN 15804 (2012). Several types of EPDs exist with respect to the life cycle stages covered. The first stage is referred to as production stage, covering mandatory parts like raw material supply, transport,

manufacturing and associated processes (A1–A3). According to EN 15804, this type of EPD is described as “cradle-to-gate”. This production phase or “cradle-to-gate” LCA consideration is addressed within this research study.

All extracted data were generated based on this production phase, hence comprising the modules transport, manufacturing and associated processes. Specifics of the data sources are further outlined in section 3.2.

The use as well as the disposal phase are not considered within this research study, as the required data are currently unavailable or rather non-existent.

2.2. CFRP reinforcement for concrete structures

Carbon fibers currently represent the most used fiber in high-performance applications (Koumoulios et al., 2019). Apart from aerospace or the automotive industry as established areas of application, the construction industry is characterized by increasing usage of CFRP reinforcement in concrete structures. This high-performance reinforcement (with a tensile strength of up to 3000 MPa and beyond) can be purposefully arranged in regions of a building structure with high occurring tensile stresses. Conventional steel reinforcement has a comparatively lower yield strength of 550 MPa (in Austria). Additionally, serviceability often represents the governing criterion in the design of FRP reinforced structures. The favorable high stiffness characteristics distinguish carbon fibers from other fiber types as basalt, aramid or glass fibers, which show lesser mechanical performance. As non-metallic reinforcement, CFRP is not characterized by electrochemical corrosiveness. Nonetheless, several environmental and mechanical factors may affect the durability of CFRP (Belarbi et al., 2019).

CFRP reinforcement is produced by bundling uni-directional filaments of carbon fibers and bonding with different types of matrices such as epoxy resin, styrene-butadiene or vinyl ester resin to form reinforcement elements. The thermoset epoxy resin is addressed within this research study. Thermosetting resins are irreversibly hardened by curing when undergoing a molecular cross-linking process. They offer a high thermal stability, good rigidity and hardness (Schumann et al., 2018).

The variety of CFRP reinforcement ranges from textile reinforcement with different mesh sizes and reinforcement ratios to extruded reinforcement as bars and tendons with different diameters and laminates with varying dimensions. External CFRP plates or sheets are widely used as post-strengthening of reinforced concrete structures. A common production method of CFRP textile reinforcement is warp knitting, subsequent impregnation and curing, whereas, rods and lamellas are produced by pultrusion, at which material is pulled through tooling devices and resin baths to merge, shape and cure the resulting CFRP composite (Gudonis et al., 2014). The surface of CFRP reinforcement elements can be plain, sand-coated or otherwise processed, leading to varying bond properties of CFRP reinforcement in concrete structures (Suppanz and Kromoser, 2020). A major issue in designing CFRP reinforced concrete structures is the bond behavior between reinforcement and concrete. Investigations on textile-reinforced concrete members have shown a so-called splitting failure-mode, resulting out of higher bond-stresses (Preinstorfer and Kollegger, 2020). Fig. 1 shows a CFRP laminate, rod and grid, at which the rod is additionally wrapped with a strand to achieve favorable bond properties.

Due to the absence of approved European design codes, the application of FRP as reinforcement in concrete structures is currently still narrowed to a certain extent. Achievements of CFRP reinforcement applied in practice proving its applicability as well as increasing research activities attempt to counter this issue. The

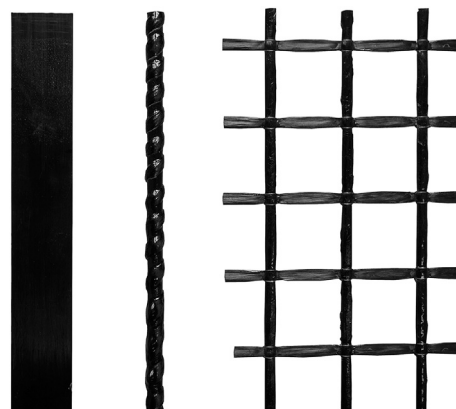


Fig. 1. CFRP reinforcement products from left to right: Laminate, rod with wrapping and grid. (Nadine Stoiber)

often claimed environmental advantage of CFRP reinforcement in comparison to conventional steel reinforcement within the related community is matter of investigation within this research study and could enhance further research or rather application of this high-performance reinforcement.

Regarding high-performance materials such as FRP reinforcement, cost performance is important to be considered. In this context the recent successful research study by Cadenazzi et al. (2020) has to be mentioned, who conducted a Life Cycle Cost analysis of alternative reinforcement bars for prestressed and reinforced concrete of a bridge design in Florida. Regarding long term implications, they mention that FRP reinforced prestressed and reinforced bridges depict comparative environmental as well as economic benefits. Furthermore the work of Grace et al. (2012) is mentioned, who found CFRP reinforced concrete highway bridges to be more cost-effective over lifetime in comparison to traditional steel reinforced bridges.

3. Method and materials

3.1. Cradle-to-gate LCA

A cradle-to-gate LCA is conducted, assessing the environmental impact of the production phase of CFRP reinforcement. Calculations were executed in compliance with relevant standardization as outlined in the theoretical framework section. The authors refrained from consulting LCA software tools on behalf of preserving maximum transparency regarding simple reproducibility of the study, information on environmental data availability as well as the identification process of impact category parameters. In general, the consultation of LCA software is conducive when processing input and output data of a considered material from scratch. As declared units (short: DU) 1 kg of the particular product (epoxy resin, carbon fiber, CFRP reinforcement, steel product) and one cubic meter of concrete are determined.

The CFRP products consist of epoxy resin and carbon fibers, at which the respective shares are outlined in section 3.2.1 and 3.2.2. The declared units are displayed in Fig. 2.

Subsequently, category impact parameters are evaluated and appraised. They intend to serve as design values allowing for a preliminary estimation of the environmental performance of CFRP reinforcement. On the one hand, the approach is to independently calculate impact categories in case necessary LCI data (European average) are provided. The LCIA method of choice is Institute of Environmental Sciences, Leiden University (2016). Characterization factors are adopted from the publicly accessible database CML-

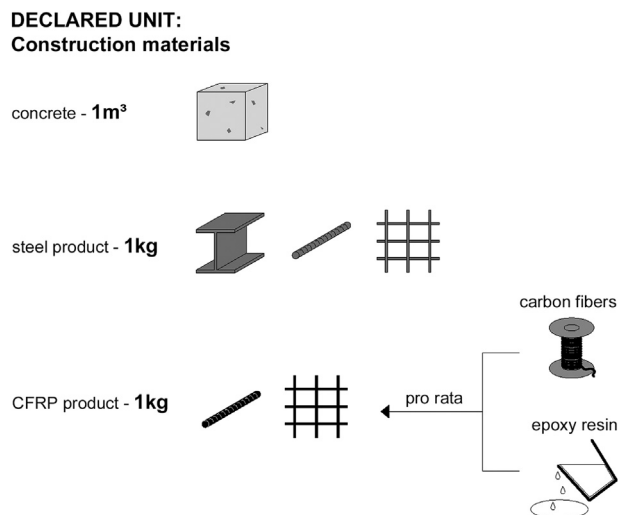


Fig. 2. Declared units of the construction materials.

IA (August 2016) of the Institute of Environmental Sciences, Leiden University. On the other hand, the approach is to adopt reliable impact category parameters from scientific information sources and EPDs. Ideally, both approaches are combined and independently calculated impact categories are checked for validity by a comparison with values extracted from EPDs. It has to be noted that LCIA methods and respective characterization factors undergo continuous reappraisal. Due to this, discrepancies can arise when aggregating environmental data from various sources. However, data sources within this research study have been carried out in compliance with the same underlying standardization. In addition, the updates of the characterization factors of CML-IA have been assessed and found to be negligible along the different consulted versions; they do not show any relevant effects on the presented results.

The parameters describing environmental impact as referred to in this paper include the global warming potential (short: GWP) in kg CO₂ equivalent, the acidification potential for soil and water (short: AP) in kg SO₂ equivalent and the abiotic depletion potential of fossil fuels (short: ADPf) in MJ. Reasons for this choice comprise data availability on the one hand and tangibility for the reader on the other hand. The authors consciously refrained from weighting or prioritizing the selected impact categories and from crediting these with environmental importance. In public and recent studies the GWP is often attached with great environmental significance, at the same time eclipsing other relevant impact category parameters. Thus, this study intends to preserve maximum neutrality and unbiasedness amongst the different impact category parameters. Initially, the authors also intended to include the ozone depletion potential (short: ODP) in kg CFC11 equivalent within this study. However, the values of the ODP as calculated and extracted from varying data sources showed inconsistent results. Due to limited information provided in EPDs and LCI data, the reason for these discrepancies couldn't be comprehended, which is why the authors finally decided to exclude it from this study.

Summing up, the underlying equation of the calculation procedure of the conducted cradle-to-gate LCA is outlined in equation (1).

$$\text{Impact Category}_{\text{construction material}} = \text{LCI Data}_{\text{construction material}} * \text{Characterization Factor} \quad (1)$$

$\text{Impact Category}_{\text{construction material}}$... represents environmental

impact parameters according to [ÖNORM EN 15804 \(2012\)](#). The Global Warming Potential (GWP) in kg CO₂ equivalents, the Acidification Potential (AP) in kg SO₂ equivalents as well as the Abiotic Depletion Potential of fossil fuels (ADPf) in MJ is calculated.

$\text{LCI Data}_{\text{construction material}}$... display the consulted cradle-to-gate input-output data of a respective considered product, e.g. g CO₂ or g Methane emissions per declared unit of construction material. The declared unit of reinforcement and epoxy resin is 1 kg and the declared unit of concrete is 1 m³ as displayed in [Fig. 2](#).

$\text{Characterization Factors}$... are retrieved from the publicly accessible database CML-IA (August 2016) of the Institute of Environmental Sciences, Leiden University (2016) as mentioned above. Via a multiplication of characterization factors.

3.2. Data sources and impact category evaluation

3.2.1. Carbon fibers

Carbon fibers are commonly produced from polyacrylonitrile (short: PAN) fibers. This chemical precursor of carbon fibers is obtained from crude oil. Process steps in the manufacturing of carbon fibers include stabilization, carbonation, surface treatment and sizing agent application. The degradation rate from PAN to carbon fiber varies from 41 to 55% amongst scientific literature as outlined by [Hohmann \(2018\)](#). For example, [Das \(2011\)](#) assumed a yield factor of 48%. The variations of the degradation rate from PAN to carbon fiber result from different system boundaries, polymeric precursors and varying process parameters like process temperature or retention time. The production of carbon fibers accounts for the predominant portion of the environmental impact of CFRP products, as further stated by [Hohmann \(2018\)](#). The manufacturing base (Europe, USA, etc.) with regard to energy supply (renewable energy, coal, gas, nuclear energy, etc.) is the most influential factor regarding the assessment of its environmental impact.

The mean values of the environmental impact categories per kg carbon fiber are extracted from the dissertation of [Hohmann \(2018\)](#) and listed in [Table 1](#). For further details regarding specifics of the LCA modules A1-A3, e.g. transport distances, it is referred to the considered dissertation of [Hohmann \(2018\)](#). The extracted values refer to [CML-IA \(2016, January\)](#) as LCIA method, the process parameters of Europe as production base, an electrical energy form, a power mix and a material yield from PAN to carbon fiber of 55%. These impact categories are compared with the results of an independent calculation, in which input and output data of PAN from the European Platform on Life Cycle Assessment (short: EPLCA) were consulted ([European Commission, 2012](#)). The European reference Life Cycle Database (short: ELCD) has been discontinued, nevertheless the dataset of PAN represents the most current publicly available dataset of its kind. The dataset (UUID: db00901a-338f-11dd-bd11-0800200c9a66) was developed by thinkstep, former PE International. The results of this dataset are aligned to a degradation rate from PAN to carbon fiber of 50%, which is stated by [Hohmann \(2018\)](#) to represent an average production scenario of carbon fibers. The values of the independent calculation are also listed in [Table 1](#). The comparatively lower values of the calculation are plausible, as no influence of a production process from PAN to carbon fibers and thus required energy was considered. However, the calculated AP is much lower than the value of [Hohmann \(2018\)](#) that shows significant differences between the minimum and maximum value in comparison to other impact categories. [Hohmann \(2018\)](#) gives the different reaction equations of the combustion of ammonia and hydrocyanic acid, which were consulted for the identification of the maximum respectively minimum value of the AP, as reason for these differences. Additionally, the GWP per kg carbon fiber of 26.4 kg CO₂ eq as outlined in [Table 1](#), correlates with an estimation of [Das \(2011\)](#) in the context of

Table 1
Impact categories per kg carbon fiber and epoxy resin.

CFRP component	Data source	GWP [kgCO ₂ eq]	AP [kgSO ₂ eq]	ADPf [MJ]
Carbon fiber	Dissertation by Hohmann (2018)	26.4	3.2E-1	404
	LCI data by ELCD (European Commission, 2012)	11.4	4.0E-2	265
Epoxy resin	LCI data by PlasticsEurope (2005)	8.6	2.1E-2	130
	EPD by IBU ^a (Institut Bauen und Umwelt e.V., 2015)	5.8	1.2E-2	117

^a Reactive resin based on epoxy resin, unfilled/solvent-free with low content of reactive diluent.

automotive industry of a GWP of 31.0 kg CO₂ eq per kg carbon fiber. The values of [Hohmann \(2018\)](#) are used for subsequent assessment of the environmental impact of CFRP reinforcement within this research study.

In this context, the Eco Impact Calculator by the European Composites Industry Association ([EuCIA, 2020](#)) has to be noted, which represents a publicly available cradle-to-gate LCA tool for composites. The category of climate change of the Eco Impact Calculator gives a prior value of 39.0 or rather an updated one of 49.1 kg CO₂ eq per kg carbon fiber, which represent higher values than the maximum GWP of [Hohmann \(2018\)](#). Once more, this circumstance mirrors the need for a continuing assessment, a higher level of transparency and comparability of the environmental performance of carbon fiber production.

3.2.2. Epoxy resin

Epoxy resins are classified as thermosetting resins, which react with the addition of epoxy hardener products. These are numerous and vary widely in their precise composition. This variety leads to differing LCI data of thermosetting resins based on epoxy resin and impedes the correct data choice regarding the particular application. Therefore, as reasonable approximation it is recommended by [Boustead \(2005\)](#) to assume the considered product, in this case the matrix material of CFRP reinforcement, as entirely made up of epoxy resin.

The results of an aggregation of cradle-to-gate LCI data from liquid epoxy resin (short: LER) produced from epichlorhydrin and bisphenol A (European production industry average) are presented in [Table 1](#). The publicly available and most recent cradle-to-gate LCI data of LER were consulted from Institut Bauen und Umwelt (short: IBU). The calculated impact categories are compared with the values of the EPD of unfilled reactive resin based on epoxy resin with a low content of reactive diluent by FEICA, the Association of the European Adhesive & Sealant Industry ([Institut Bauen und Umwelt e.V., 2015](#)). According to a response by FEICA, this EPD is likely to be comparable to epoxy resin used as matrix material in CFRP products. As can be seen in [Table 1](#), the independently evaluated impact categories are slightly above the values extracted from the EPD and indicate the validity of the calculation.

3.2.3. CFRP

The impact categories of carbon fibers and epoxy resin are combined in accordance to their mass fractions of the particular CFRP product, in which a distinction between rebar or rather laminate as well as CFRP mesh is made. It has to be noted that, in contrast to conventional steel reinforcement, CFRP reinforcement products are characterized by a greater bandwidth of stiffness and strength properties. The thinner the roving, the better the activation of carbon fibers embedded within the matrix material, which results in higher strength properties. Regarding the subsequent assessment of the environmental performance, the authors aimed at choosing representative tensile strength values by consulting data sheets of CFRP reinforcement products.

The safety data sheet of the carbon laminate product S&P C-Laminate by [S&P-Handels GmbH \(2016\)](#) indicates that, on average,

the product consists of 62.5% carbon fibers and 37.5% epoxy resin. The validity of these values was checked with mass fractions of CFRP rebar products by other providers. For example, the technical data sheet of the rebar product *solididan REBAR Ø8-CCE* by [solidian \(2020\)](#) indicates a carbon fiber content of 63%. The characteristic value of the tensile strength of this representative rebar product is 2100 MPa and adopted for subsequent environmental assessment. CFRP rebar is commonly produced via pultrusion. According to [Hohmann \(2018\)](#), the energy demand for pultrusion per kg is 3.1 MJ. As this value is rather small in comparison to the ADPf of 1 kg epoxy resin or rather carbon fibers, it is neglected within the assessment of the environmental performance of CFRP rebar and laminate.

Regarding CFRP meshes, the general type approval of a method to reduce crack width by the use of textile reinforcement near the surface of concrete structures by [Quinting Zementol GmbH \(2015\)](#) indicates a percentage of 55% of carbon fibers and a percentage of 45% of epoxy resin in CFRP meshes. The validity of these values was also checked with the textile reinforcement mesh product *solididan GRID Q95/95-CCE-38*, which is characterized by a fineness of the rovings in both directions of 6400 tex and a mesh size of 38 mm ([Rempel, 2018](#)). The characteristic value of the tensile strength of this representative CFRP mesh product in longitudinal direction of 2300 MPa is adopted from the respective technical data sheet by [solidian \(2017\)](#). The environmental impact of the manufacturing of CFRP meshes (warp knitting, impregnation) is not available and neglected within the scope of this assessment. However, the impact of warp knitting is expected to be even smaller than the impact of pultrusion.

The results of CFRP rebars and meshes are listed in [Table 3](#). It can be seen that, due to a higher carbon fiber content, the considered impact category parameters of CFRP rebar are slightly above the ones of CFRP meshes. The impact category values of carbon fibers as extracted from the dissertation of [Hohmann \(2018\)](#) and the independently calculated values of epoxy resin based on the LCI data by [PlasticsEurope \(2005\)](#) are consulted for subsequent environmental assessment.

3.2.4. Steel products

Steelmaking processes include basic oxygen steelmaking (short: BOS), in which pig iron from blast furnace (short: BF) and steel scrap is made into steel. Another manufacturing method is electric arc furnace (short: EAF), melting steel scrap to produce liquid steel. According to the [World Steel Association \(2019\)](#), about two thirds of steel in Europe have been produced by BOS in 2019. As coal is added during the production of steel by BOS, the environmental impact of EAF, which is mostly based on steel scrap and electricity, is considered to be comparatively lower.

Regarding the impact category parameters of steel reinforcement, as an initial step, several EPDs were consulted and opposed. The EPDs of conventional steel reinforcement listed in [Table 2](#) are currently valid and carried out in accordance with [ISO 14025 \(2006\)](#) and [ÖNORM EN 15804 \(2012\)](#). The considered EPDs are listed by means of an increasing GWP. The manufacturing process seems to have a significant influence on the impact categories, as the values

of steel reinforcement produced in BOS/BF are comparatively higher than those produced in EAF. Although BOS represents the major share in steelmaking in Europe as stated by the [World Steel Association \(2019\)](#), EPDs with EAF as manufacturing process are found more frequently. The differences between steel rebar and mesh are found to be marginal.

Subsequently, the impact category parameters are independently calculated based on up-to-date cradle-to-gate LCI data provided by the [World Steel Association \(2019\)](#), which can be requested online and represents European average data. It includes steel production from the integrated route (BOS/BF) based on iron ore and steel scrap as well as the EAF route mainly based on steel scrap. The data consulted within this research study does not consider a burden for scrap input or a credit for end-of-life recycling. In this context, it has to be mentioned that steel is especially characterized by recyclability, as steel scrap recycled from steel products at the end of their lives represents an essential input to the steelmaking process. The [World Steel Association \(2019\)](#) also provides LCI data of steel products including a recycling rate of 85%. The results for 1 kg steel rebar representing steel reinforcement and 1 kg steel sections, excluding recycling, are listed in [Table 3](#). The impact categories GWP and AP of 1 kg hot-dip galvanized steel were adopted from [World Steel Association \(2018\)](#), at which the ADPf was aligned to the independently calculated results of the ADPf of 1 kg steel section.

3.2.5. Concrete

The publicly available EPDs by InformationsZentrum Beton GmbH ([Becke et al., 2014](#)) are consulted to integrate the environmental impact of concrete used within the subsequent comparison of a conventionally and CFRP reinforced concrete bridge design. The considered cradle-to-gate impact categories of 1 m³ of an average standard composition of concrete in Germany are extracted, at which six different concrete qualities are given (C20/25, C25/30, C30/37, C35/45, C45/55 and C50/60). Average transport distances of raw materials to a manufacturing plant include 30–60 km for aggregate, around 180 km for cement and around 100 km for additives and fly ash. The production of concrete's binder cement accounts for the major portion of the global environmental impact, at which the amount of cement clinker has a significant influence on the results. When comparing the impact categories of the EPDs provided by InformationsZentrum Beton GmbH ([Becke et al., 2014](#)), a steady dependence between these becomes apparent. It follows that an extrapolation of the concrete quality C70/80 seems to represent an acceptable approach to estimate its impact categories. The impact category values are listed in [Table 3](#) and are consulted for the subsequent environmental assessment. The authors consciously refrained from assessing the environmental impact of individual concrete mixtures as used for the bridge examples in section 4.2, as the intention of this research study is to incorporate universally valid design values as provided by the InformationsZentrum Beton GmbH ([Becke et al., 2014](#)).

In this context, a study by [Purnell and Black \(2012\)](#) has to be mentioned, which illustrates the variation of the embodied carbon in accordance with the mean cube strength. Broadly speaking, the

results indicate that less favorable relations are given for normal and ultra-high strength concrete in comparison to high-strength concrete. The study shows that different concrete qualities might not feature the same trend in environmental degradation, provided an efficient application is given. This study points out the need for EPDs representing average mixtures of high strength and ultra-high strength EPDs.

4. Results and discussion

The resulting impact category values are summarized and listed in [Table 3](#). These values are consulted for the subsequent comparison of CFRP with steel reinforcement (declared unit: kg) and the comparison of the bridges (functional unit: outlined in [Fig. 8](#)).

4.1. Comparison of CFRP and steel reinforcement

The results of the environmental impact of 1 kg steel and CFRP reinforcement are illustrated in [Fig. 3](#). The results of the environmental impact of 1 kg steel and CFRP reinforcement per MPa tensile respectively yield strength as functional unit are illustrated in [Fig. 4](#). The respective tensile strength for CFRP rebar is 2100 MPa, for CFRP textile 2300 MPa and the yield strength for steel reinforcement is 550 MPa. It is apparent that CFRP reinforcement shows a much higher environmental impact than steel reinforcement amongst all impact categories. The gap is smaller when considering the impact per tensile respectively yield strength as in [Fig. 4](#), but one might still, for example, use two times as much steel reinforcement as CFRP reinforcement in order to align the GWP. [Fig. 5](#) shows the environmental impact per tensile or rather yield strength as well as per density unit, at which the density of steel is 7850 kg/m³ and the assumed density of CFRP reinforcement is 1500 kg/m³ according to the data sheet by [solidian \(2020\)](#). The results adjusted in accordance with the density of the reinforcement types is illustrated in [Fig. 5](#) and are comparatively more in favor of CFRP reinforcement regarding GWP and ADPf.

It can be noticed that the AP of CFRP reinforcement is especially high in comparison to steel reinforcement. As already noted in section 3.2.1, the AP is characterized by a comparatively great difference between maximum and minimum value. [Hohmann \(2018\)](#) gives the different reaction equations of the combustion of ammonia and hydrocyanic acid, which were consulted for the identification of the maximum respectively minimum value of the AP, as reason for these differences. This mirrors the fluctuations in the environmental assessment of CFRP reinforcement, which are associated with the varying or rather intransparent production process parameters.

Partial safety factors are not taken into account within this assessment, which would result in an even higher difference of the environmental impact in favor of steel reinforcement. As elaborated by [Rempel \(2018\)](#), the recommended partial safety factor of textile reinforcement γ_t is 1.3. According to ONORM EN 1992-1-1:2015-02, 2015, the partial safety factor γ_s of steel reinforcement is 1.15, in case of the permanent and temporary design situation regarding load bearing capacity.

Table 2

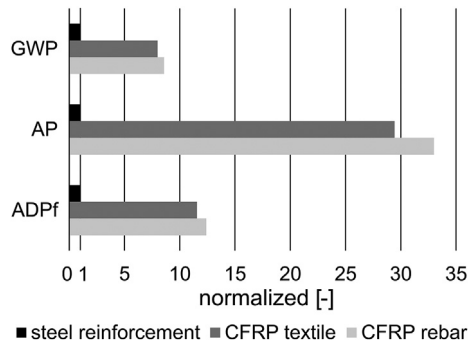
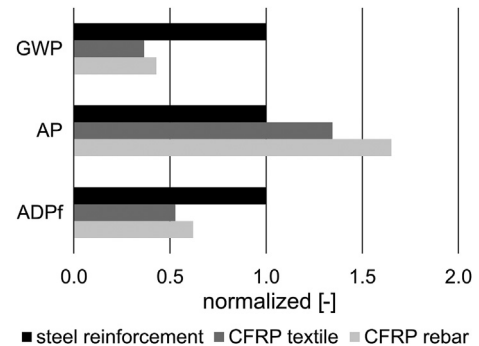
Impact categories per kilogram steel reinforcement: List of EPDs from different regions in Europe ranked by CO₂ equivalents.

Programme operator	Region	Production Process ^a	GWP [kgCO ₂ eq]	AP [kgSO ₂ eq]	ADPf [MJ]
ift Rosenheim GmbH (2018)	Germany	EAF mainly	0.2	6.7E-4	4.0
EPD International AB (2015)	Sweden	EAF	0.4	7.8E-4	2.0
BRE Global (2019)	Spain	EAF	0.5	1.5E-3	5.8
BRE Global (2018a)	Turkey	EAF	1.0	4.4E-3	11.8
BRE Global (2018b)	Ukraine	BOS/BF	2.6	2.9E-3	24.9

^a EAF ... electric arc furnace, BOS ... basic oxygen steelmaking, BF ... blast furnace.

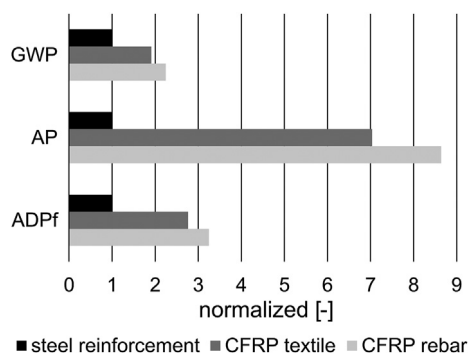
Table 3Impact categories per kg steel or CFRP product and per m³ concrete quality.

Material	Product	DU	Data source	GWP [kgCO ₂ eq]	AP [kgSO ₂ eq]	ADPf [MJ]
CFRP	rebar	kg	proceed to Table 1	19.7	2.1E-1	301
	textile			18.4	1.9E-1	281
Steel	reinforcement	kg	World Steel Association (2019)	2.3	6.3E-3	24.1
	section			1.5	3.5E-3	16.7
Concrete	hot-dip galvanized ^a	m ³	InformationsZentrum Beton GmbH (Becke et al., 2014)	2.8	6.4E-3	27.3
	C30/37			232	3.2E-1	845
	C50/60			335	4.5E-1	1230
	C70/85 ^b			431	5.7E-1	1567

^a Adopted or rather aligned in accordance with World Steel Association (2018)^b Extrapolated.**Fig. 3.** Comparison of the environmental impact of CFRP and steel reinforcement per declared unit kg (normalized values).**Fig. 5.** Comparison of the environmental impact of CFRP and steel reinforcement per MPa characteristic tensile respectively yield strength as well as per density unit (normalized values).

4.2. Comparison of bridge examples

The comparison of the impact categories in Fig. 4 constitutes that CFRP reinforcement is characterized by a higher environmental impact than conventional steel reinforcement in relation to their strength properties. To take other characteristics of CFRP reinforcement, such as non-corrosiveness and thus the possibility to build light-weight concrete structures into account, a comparison of existing bridges regarding their environmental performance was conducted for the cradle-to-gate phase as exemplary application of the presented data. The superstructures of three bridges of different methods of construction as illustrated in Fig. 6 are contrasted. The first bridge *B1-CCB* (bridge example 1: carbon concrete

**Fig. 4.** Comparison of the environmental impact of CFRP and steel reinforcement per declared unit kg as well as per MPa characteristic tensile respectively yield strength (normalized values).

bridge) represents a textile-reinforced carbon concrete bridge, which was built in 2015 in Albstadt, Germany. The second bridge example *B2-RCB* (bridge example 2: reinforced concrete bridge) is a reinforced concrete bridge with conventional steel reinforcement and the third bridge *B3-SB* (bridge example 3: mild steel bridge) is made of mild steel. Both of the two last-named bridges are located in Vienna, Austria and were built in 2009 (*B2-RCB*) or rather 1999 (*B3-SB*).

Subsequently, the boundary conditions of the bridges are outlined and the masses of the superstructures are determined. Finally, the environmental impacts of all bridges are evaluated and the results are discussed in consideration of the boundary conditions.

4.2.1. Boundary conditions

All relevant boundary conditions of the compared bridge examples are listed in Table 4. The imposed loads of the bridges were either extracted from respective publications (Helbig et al., 2016) or original static calculations. *B1-CCB* was calculated in accordance with DIN EN 1991-2 (2010) with an imposed load of 4.66 kN/m². Due to the lack of technical approvals for carbon concrete, a series of experiments had been conducted to receive an approval in this individual case (German: Zulassung im Einzelfall, short: ZIE). The consulted documentation of the static calculation of *B2-RCB* outlines the underlying assumptions for this calculation: An effective width of 3.5 m, an imposed load of 5.0 kN/m² according to ÖNORM EN 1991-2 (2004) as well as an additional safety load of 1.7 kN/m² in case of an additional asphalt layer applied later on. The static analysis of the reinforced concrete bridge was conducted in accordance with ÖNORM EN 1992-2 (2007).



Fig. 6. Consulted bridge examples from top to bottom: B1-CCB (solidian GmbH), B2-RCB and B3-SB (Mathias Hammerl).

B3-SB has already been built in 1999 before the implementation of the Eurocodes. Therefore, the underlying national design code for imposed loads was *ÖNORM B 4002 (1975)*. For bridge class II as in this case, an imposed load of 400 kg/m^2 was adopted. The structural analysis was carried out in accordance with *ÖNORM B 4600 (n.d.)* series and *ÖNORM B 4602 (1970)*.

The support span of all bridges slightly varies around 15 m and the effective width ranges from 2.5 to 3.0 m. Due to these minor differences, a comparability of the environmental performance of these bridges is valid.

4.2.2. Quantity survey

Based on the original construction plans provided by the Municipal Department 29 in Vienna, which is responsible for bridge construction and maintenance in Vienna, a precise determination of the masses of the bridges B2-RCB and B3-SB was feasible. The masses of B1-CCB were provided by the bridge designers from Helbig et al. (2016). In Fig. 7, the different cross sections are illustrated. B1-CCB and B2-RCB represent single-span

girder bridges with a U-shaped cross section. For B1-CCB, a concrete quality of C70/85 was used for the concrete deck as well as the concrete walls. As carbon textile reinforcement solidian *GRID Q95/95-CCE-38* by solidian (2017) was used and the railing is made out of hot-dip galvanized steel. The deck of B2-RCB is made of concrete C50/60 and the walls are made of concrete C30/37. Conventional steel reinforcement with a yield strength of 550 MPa was used for B2-RCB. The third bridge B3-SB represents a single-span girder bridge made of mild steel. The primary structure is made of HEA 360, the secondary structure is made of IPE 200 and the railing is made of hollow and flat steel sections. The determined masses of all three bridges are summarized in Table 5.

4.2.3. Analysis of the environmental impact

The environmental impact of the three bridges is compared within this section. The functional unit per bridge comprises the mass of reinforcement and railing in kg, the volume of concrete in m^3 and the underlying function of it as a pedestrian bridge with a span of approximately 15 m and an effective width of approximately 3 m. The support span as well as the width of the bridges illustrated in Table 4 deviate only slightly from each other, which allows for the comparability of the results. The bridges' static system is a single span girder. The functional unit is displayed in Fig. 8.

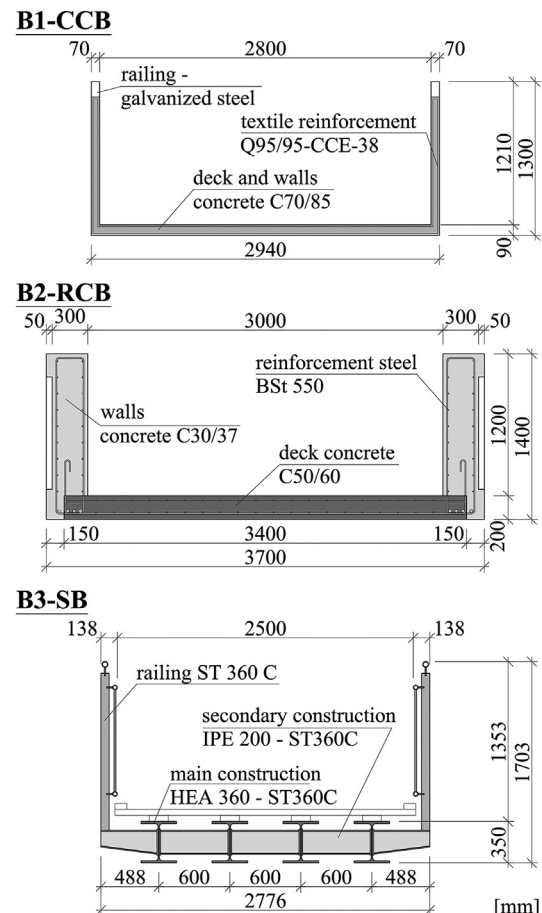


Fig. 7. Cross sections of consulted bridge examples from top to bottom: B1-CCB, B2-RCB and B3-SB (dimensions in mm).

Table 4
Boundary conditions of the consulted bridge examples B1-CCB, B2-RCB and B3-SB.

	Bridge 1	Bridge 2	Bridge 3
Abbreviation	B1-CCB	B2-RCB	B3-SB
Type	carbon concrete	reinforced concrete	mild steel
Year	2015	2009	1999
Imposed load [kN/m^2]	4.7 ^a	5.0 (6.7) ^b	4.0 ^b
Support span [m]	15.6 ^c	15.0 ^c	14.6 ^c
Effective width [m]	2.8 ^c	3.0 ^c	2.5 ^c

^a Value adopted from Helbig et al. (2016).

^b Value extracted from original static calculation.

^c Dimensions extracted from original construction plans.

FUNCTIONAL UNIT: Pedestrian bridge

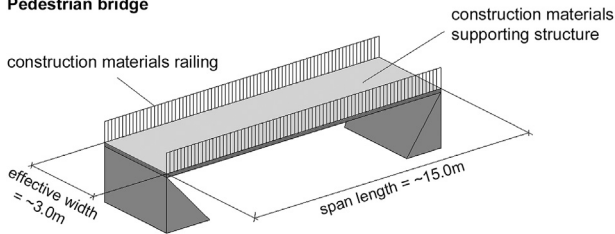


Fig. 8. Functional unit - superstructure of a pedestrian bridge with a span length of approximately 15.0 m and an effective width of approximately 3.0 m.

Table 5

Quantity survey of the consulted bridge examples B1-CCB, B2-RCB and B3-SB.

Bridge type	Construction element	Value	Unit
B1-CCB	deck and walls C70/85	6.16 (14.88)	m ³ (t)
	textile reinforcement ^a	328.60 (0.21)	m ² (t)
	railing galvanized steel	0.83	t
B2-RCB	deck C50/60	10.60 (25.44)	m ³ (t)
	walls C30/37	12.48 (29.95)	m ³ (t)
	steel reinforcement BSt 550	2.93	t
B3-SB	primary structure ST360C	6.55	t
	secondary structure ST360C	1.43	t
	railing ST360C	0.77	t

^a Solidian GRID Q95/95-CCE-38 (2017).

Summing up, the underlying equation of the calculation procedure of the conducted cradle-to-gate LCA of the three bridge types is outlined in following equation (2).

$$\text{Environmental Impact}_{\text{bridge type}} = \text{Impact Category}_{\text{concrete}} * \text{Volume}_{\text{concrete}} + \text{Impact Category}_{\text{steel product}} * \text{Mass}_{\text{steel product}} + \text{Impact Category}_{\text{CFRP reinforcement}} * \text{Mass}_{\text{CFRP reinforcement}} \quad (2)$$

$\text{Environmental Impact}_{\text{bridge type}}$... represents the accumulated environmental impact per bridge, more precisely per functional unit according to Fig. 8.

$\text{Impact Category}_{\text{concrete}}$, $\text{Impact Category}_{\text{steel product}}$ and $\text{Impact Category}_{\text{CFRP reinforcement}}$... are calculated according to equation (1).

$\text{Volume}_{\text{concrete}}$, $\text{Mass}_{\text{steel product}}$ and $\text{Mass}_{\text{CFRP reinforcement}}$... are determined in section 4.2.2.

The environmental impact categories of the three consulted bridges, in dependence on the determined masses as listed in Table 5 and the environmental data as elaborated in Table 3, are illustrated in Fig. 9. B1-CCB indicates a significant reduction of the GWP as opposed to the two conventional bridge types. B2-RCB and B3-SB show a GWP of almost 1.5 times as high as the one of B1-CCB. However, B1-CCB has the highest AP. This can be explained by considering the AP of carbon fibers, which is about 50 times higher than the one of steel reinforcement. The comparative reduction of concrete volume of B1-CCB does not result in a compensation of the AP. Regarding the ADPf, B1-CCB and B2-RCB show almost the same environmental impact. It is noticeable that the ADPf of B3-SB is about 1.6 times as high as the one of the two concrete bridges.

Subsequently, the impact categories undergo a deeper analysis and are broken down by the individual components of the bridges (reinforcement, concrete, mild/galvanized steel). The GWP (Fig. 10) of steel reinforcement of B2-RCB is twice as high as the one of the

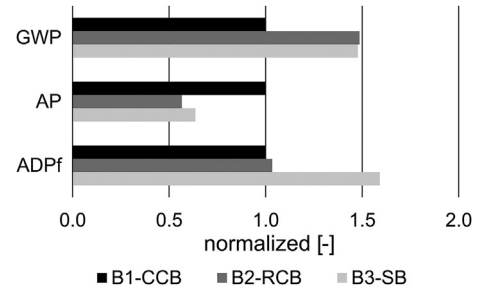


Fig. 9. Comparison of the environmental impact categories of the three consulted bridge examples.

carbon textile reinforcement of B1-CCB. Also, the GWP of concrete of B2-RCB is about twice as high as for B1-CCB. The steel reinforced bridge B2-RCB has almost the same GWP as B3-SB. These findings outline the high GWP of concrete or rather cement. Comparing the bridges' masses, it can be said that the concrete volume of B2-RCB shows the same GWP as around half of the mild steel mass of B2-SB.

Considering the AP of the three consulted bridge examples as shown in Fig. 11, it is apparent that carbon textile reinforcement shows the highest impact amongst all considered materials. The AP is about twice as high as the one of steel reinforcement of B2-RCB and about one third higher than the whole mild steel mass used for B3-SB.

Fig. 12 shows that the ADPf for the used mild steel mass in B3-SB is about 1.5 times as high as the one of B1-CCB or rather B2-RCB. The steel reinforcement of B2-RCB has a slightly higher ADPf than the carbon reinforcement of B1-CCB.

Considering Figs. 10 and Figs. 12, it can be seen that the galva-

nized steel used for the railing of B1-CCB has a significant share in the GWP and ADPf. Therefore, Fig. 13 illustrates an adaption of Fig. 9, at which the railing of B1-CCB made of galvanized steel is replaced with CFRP reinforced concrete of the same thickness as the walls of the U-shaped cross section of B1-CCB. This would lead to a reduction of galvanized steel of 830 kg and an increase of 0.56 m³ concrete C70/85 and 10.3 kg of carbon textile reinforcement. The AP of B1-CCB is still about 40% higher than the one of B2-

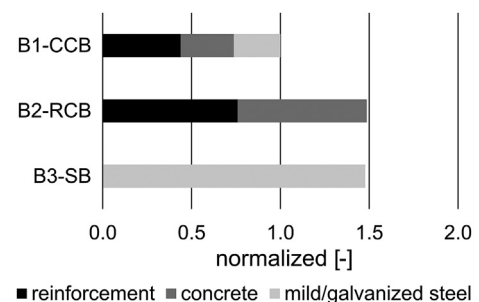


Fig. 10. GWP broken down by the individual components of the three consulted bridge examples.

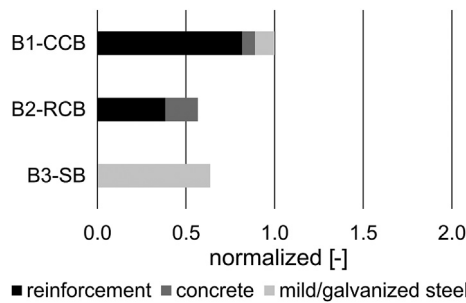


Fig. 11. AP broken down by the individual components of the three consulted bridge examples.

RCB and B3-SB, but the GWP as well as the ADPf is comparatively lower than illustrated in Fig. 9.

By now, recyclability of steel hasn't been taken into account when assessing the environmental impact of the CFRP reinforced and the conventionally reinforced bridge or rather the mild steel bridge. Fig. 14 shows the comparison of the consulted bridge examples, at which a recycling rate of steel products (reinforcement, mild steel, galvanized steel) of 85% in accordance with the available data by World Steel Association (2019) was considered. Overall, the results indicate only minor deviations from the results excluding recycling in Fig. 9. However, the ADPf of B2-RCB is lower than the one of B1-CCB.

Regarding the undertaken comparisons of the environmental impact of the bridge examples, it is inevitable to take the given boundary conditions of the bridge examples into consideration (Table 4). According to Helbig et al. (2016), the imposed load adopted for the calculation of B1-CCB was 4.66 kN/m². B2-RCB has the comparatively highest live load and for the static calculation a width of 3.5 m was assumed. The mild steel bridge B3-SB has the comparatively lowest imposed load and width. Taking these slightly varying boundary conditions and a theoretical alignment of these amongst the considered bridge examples into account, it can be stated that the category impact values of the conventionally reinforced concrete bridge (B2-RCB) tend to decrease in comparison to the carbon concrete bridge (B1-CCB). The values of B1-CCB as well as the values of B2-RCB tend to decrease as opposed to the steel bridge (B3-SB).

When comparing the environmental impact of varying bridge construction types, also their intrinsic disadvantages and advantages should be taken into account, at which the most significant ones are discussed. Considering the structural design of the three considered single-span girder bridges, the B3-SB consists of a comparatively simple cross section layout, however, U-shaped concrete cross sections are also well-established within civil engineering. An advantage of the concrete cross section of B1-CCB and B2-RCB is that the walls can be used as structural elements as well as as railing, hence combining multiple functions. B1-CCB and B3-SB

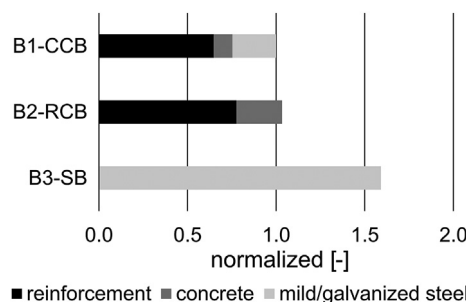


Fig. 12. ADPf broken down by the individual components of the three consulted bridge examples.

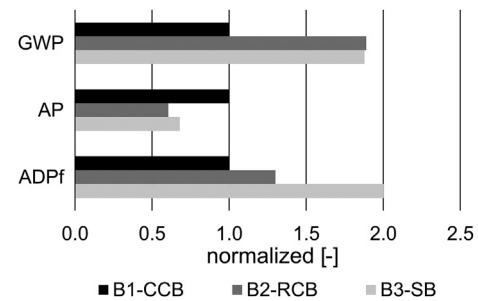


Fig. 13. Comparison of the environmental impact categories of the three consulted bridge examples by means of an adapted version of B1-CCB (railings made of galvanized steel are replaced with CFRP reinforced concrete walls).

have a high potential level of prefabrication due to their low self-weight, whereas the conventional steel reinforced concrete bridge is to be built primarily on site. According to the Austrian Federal Railways, the life cycle of a steel bridge is approximately 100 year or rather 70 years in case of a concrete bridge. By using non-corrosive CFRP, the service life of the respective structure, which is exposed to harsh environment (Belarbi et al., 2019), can be extended. For a more extensive application of CFRP in construction, investigations regarding its durability properties are necessary.

Finally, it has to be noted that only the superstructure of the bridges was considered within the assessment of the environmental performance. An inclusion of the bridges' substructures is expected to lead to a result even more in favor of the carbon concrete bridge, as less self weight requires smaller foundations. The use as well as the disposal phase were not considered within this evaluation.

5. Conclusion

This study elaborated on the environmental performance of CFRP reinforcement in concrete structures, at which a cradle-to-gate LCA was conducted. Design values for the estimation of the environmental impact represented by three impact categories, including the global warming potential (GWP), acidification potential (AP) and abiotic depletion of fossil resources (ADPf), are provided. The evaluations are built on the current state of knowledge and standardization as well as the availability of necessary environmental data. Maximum transparency of the identification process of the impact categories was set as top priority.

The opposition of CFRP reinforcement and conventional steel reinforcement in regard to the respective yield or rather tensile strength outlined the comparatively high environmental impact of CFRP reinforcement. These results indicate steel reinforcement as preferred reinforcement choice in concrete structures when optimizing the environmental performance. Subsequent investigations of the environmental performance of a carbon concrete, reinforced

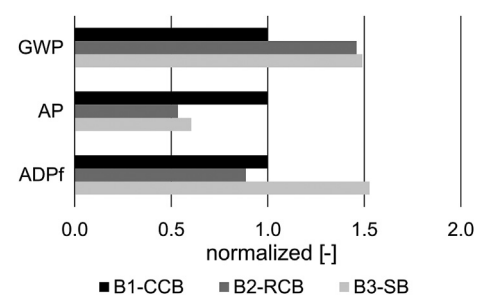


Fig. 14. Comparison of the environmental impact categories of the three consulted bridge examples with consideration of the steel products' recycling rate of 85%.

concrete and mild steel bridge showed otherwise, at which the lowest GWP and ADPf is associated with the CFRP reinforced bridge. However, the AP of the carbon concrete bridge is comparatively higher, due to the high AP of carbon fibers.

This paper conducted a cradle-to-gate LCA and took the product phase of materials into account. Regarding the use phase of carbon concrete, the authors expect results in favor of CFRP reinforcement due to a comparatively lower weight and thus less transport emissions. Another advantageous characteristic of CFRP reinforcement is non-corrosiveness, which is currently expected to result in a comparatively longer service life and less expenses in maintenance. A notable topic of great importance regarding the disposal phase of a product is recyclability, which has to be especially addressed within the context of materials' environmental performance. The consideration of a recycling rate of steel products of 85% within this paper did not lead to a significant change in the results. On a case-by-case basis, one might expect results more in favor of steel reinforcement when consulting material out of an especially optimized steelmaking process with little environmental impact (compare with values extracted from EPDs in Table 2). As for CFRP reinforcement, recyclability represents a currently flourishing topic of research. For example, Merli et al. (2020) outlines this research activity via a conducted literature review about recycled fibers in reinforced concrete, including CFRP. Kimm et al. (2018) found that though the separation of concrete and textile reinforcement is feasible, however, it can hardly be reused for the original application as reinforcement. Construction materials made of recycled short fibers are currently developed and investigated by Kimm et al. (2018). The authors suggest or rather expect this research progress in recycling process optimization and manufacturing method development of recycled carbon fibers to continue in the future. An extension of the conducted evaluation of environmental impacts to a cradle-to-gate LCA with options, considering end-of-life as well as the construction phase, is suggested.

To sum up, this study shed light on the issue of environmental impact of CFRP reinforcement in concrete structures, which is often carelessly assumed to be small within the related community. The results indicate CFRP reinforcement to be a justified environmentally friendly alternative to steel reinforcement in case of the considered bridge design, provided that a maximum utilization of its strength and stiffness properties is aimed at and optimally implemented within the design process. One possible approach to improve this utilization is prestressing of CFRP reinforced structures. The positive effect on the serviceability of prestressing CFRP rebars in CFRP reinforced beams was investigated by Hammerl and Kromoser (2020) and showed promising results. It has to be noted that a generally valid statement regarding the environmental performance of CFRP reinforced concrete structures can not be drawn from the considered exemplary application within this study. Regarding future research, it is crucial to continue the environmental assessment of CFRP reinforcement in concrete structures, as necessary available environmental data is subject to change and gets updated or rather more transparent over time.

Credit author contribution statement

Nadine Stoiber: Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing, Visualization. **Mathias Hammerl:** Formal analysis, Writing - original draft, Writing - review & editing, Visualization. **Benjamin Kromoser:** Overarching Scientific Concept, Conceptualization, Validation, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing

financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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